

BLDC/PMSM Parameter Identification Experiment Plan

Motor Pilotix 6210 210KV, SKU 7206. Assumptions: balanced three-phase motor, non-salient rotor, $L_d = L_q = L_s$, sinusoidal PMSM-equivalent model, advertised values treated as accurate.

Quantity	Advertised value / assumption	Calculated value used in model
KV	210 rpm/V	Input value
Pole pairs p	28 magnets -> 14 pole pairs	$p = 14$
Line-to-line resistance R_{LL}	42.5 mOhm	Input value
Phase resistance R_s	Balanced star: $R_{LL} = 2 R_s$	$R_s = 21.25$ mOhm
Back-EMF / torque constant $K_e = K_t$	$K_e = 60/(2\pi \cdot KV)$	$K_e = K_t = 0.04547$ V*s/rad = 0.04547 N*m/A
Permanent magnet flux linkage ψ_{pm}	$K_t = p \cdot \psi_{pm}$	$\psi_{pm} = 3.248$ mWb

Important convention note: the K_e , K_t , and ψ_{pm} calculation uses the common simplified hobby-motor convention where the advertised KV is treated as accurate and directly converted with $K_e = 60/(2\pi \cdot KV)$, followed by $\psi_{pm} = K_e/p$. If your FOC implementation uses peak phase variables and the standard torque equation $T_e = 1.5 \cdot p \cdot \psi_{pm} \cdot i_q$, keep the convention consistent when entering ψ_{pm} and interpreting i_q .

1. Parameters that do not need experiments

The following values are calculated directly from the advertised data and should be entered as initial model parameters without a dedicated identification experiment:

- Phase resistance: $R_s = R_{LL}/2 = 42.5 \text{ mOhm} / 2 = 21.25 \text{ mOhm}$.
- Pole pairs: $p = 28/2 = 14$.
- Back-EMF / torque constant: $K_e = K_t = 60/(2\pi \cdot 210) = 0.04547 \text{ V*s/rad} = 0.04547 \text{ N*m/A}$.
- Permanent magnet flux linkage: $\psi_{pm} = K_t/p = 0.04547/14 = 3.248 \text{ mWb}$.

Temperature correction for R_s

If the winding temperature differs from the datasheet reference temperature, correct the copper resistance using:

$$R_s(T) = R_s(T_{ref}) * [1 + 0.00393 * (T - T_{ref})]$$

For simulation, store the reference temperature together with R_s . For control, update R_s online if a winding-temperature estimate is available.

2. Experiment A - Stator inductance L_s using an LCR meter

Goal: measure the line-to-line inductance L_{LL} and convert it to the per-phase stator inductance L_s . This is the only planned experiment for L_s because the datasheet does not provide inductance.

Item	Requirement
Rotor condition	Rotor mechanically locked. Repeat at several rotor angles if possible.
Connection	Measure between two motor phases, for example A-B. Leave the third phase open.
Instrument	LCR meter with low-resistance/low-inductance fixture compensation.
Frequencies	Recommended: 100 Hz, 1 kHz, and 10 kHz. Use 1 kHz as the main value unless your inverter current loop operates much lower.
Signal level	Small-signal measurement, preferably below magnetic saturation.

Calculation:

For a balanced star-connected motor, the measured line-to-line inductance is approximately twice the phase inductance:

$$L_s = L_{LL} / 2$$

Recommended reporting format:

$$L_s = \text{mean}(L_{AB}, L_{BC}, L_{CA})/2, \text{ reported together with measurement frequency, rotor lock angle, and winding temperature.}$$

Quality checks: the three line-to-line measurements should be close to each other. Large differences indicate measurement error, damaged winding, or a non-ideal saliency/slotting effect. For a non-salient simulation model, use the average value.

3. Experiment B - Split calculation for rotor inertia J and viscous friction B

Goal: identify J and B from two speed transients: one current-controlled acceleration and one passive coast-down. This avoids assuming B during the J calculation.

Mechanical model:

$$J \cdot d\omega/dt + B \cdot \omega = T_e$$

Use the motor without propeller or external load. If a shaft coupler, encoder disk, or fixture is attached, the identified J includes those attached rotating parts.

Test procedure

- 1 Remove propeller and any load that is not part of the rotor inertia to be identified.
- 2 Run FOC current control if available. Command a constant positive q-axis current i_q , for example a low safe value that produces a clean acceleration without overheating.
- 3 Record time, mechanical speed ω , and i_q during acceleration over a useful speed range. Sampling should be fast enough to compute a smooth derivative, for example 500 Hz to 2 kHz if available.
- 4 At the selected upper speed, set torque-producing current to zero or disable drive torque and record the passive coast-down $\omega(t)$. Avoid active braking.
- 5 Filter $\omega(t)$ lightly, then compute $\alpha = d\omega/dt$ for both acceleration and coast-down.

Split calculation at the same speed

At a chosen speed ω_i , evaluate the acceleration slope $\alpha_{up}(\omega_i)$ and the coast-down slope $\alpha_{down}(\omega_i)$. During acceleration:

$$J \cdot \alpha_{up} + B \cdot \omega_i = T_e$$

During coast-down:

$$J \cdot \alpha_{down} + B \cdot \omega_i = 0$$

Subtract both equations to solve inertia directly:

$$J_i = T_{e_i} / [\alpha_{up}(\omega_i) - \alpha_{down}(\omega_i)]$$

Then solve viscous friction:

$$B_i = -J_i \cdot \alpha_{down}(\omega_i) / \omega_i$$

Use $T_{e_i} = K_t \cdot i_{q,i}$ with $K_t = 0.04547 \text{ N}\cdot\text{m}/\text{A}$ under the convention of this document. If using the standard PMSM peak-current equation in your software, use $T_e = 1.5 \cdot p \cdot \psi_{pm} \cdot i_q$ consistently.

Data processing recommendations

- Perform the calculation at many speeds in the middle of the measured range, not near zero speed where Coulomb friction and quantization dominate.
- Reject outliers caused by current-loop transients, speed-estimator delay, or derivative noise.
- Report J as the mean or median of J_i and B as the mean or median of B_i .
- Repeat the test for two or three current levels. J should remain approximately constant; B may vary slightly with temperature and bearing condition.
- If B_i is not constant versus speed, the friction model is not purely viscous. Then model friction as $B \cdot \omega + T_c \cdot \text{sign}(\omega)$ if needed.

4. Measurement log template

Parameter	Value / notes
Motor	Pilotix 6210 210KV, SKU 7206
Ambient temperature	
Estimated winding temperature	
R _s used	21.25 mOhm at datasheet reference temperature
L _{AB} @ 1 kHz	
L _{BC} @ 1 kHz	
L _{CA} @ 1 kHz	
Computed L _s	$\text{mean}(L_{AB}, L_{BC}, L_{CA})/2$
Acceleration current i _q	
Speed range used for J,B fit	
Identified J	
Identified B	
Comments / anomalies	

5. Final parameter set format

Model parameter	Value
R _s	21.25 mOhm, temperature-corrected if required
L _s	From LCR experiment: $L_s = L_{LL}/2$
psi _{pm}	3.248 mWb
J	From split acceleration/coast-down calculation
B	From split acceleration/coast-down calculation

Source data used: advertised motor data from the provided Motor Pilotix 6210 210KV datasheet: KV = 210 rpm/V, internal resistance = 42.5 mOhm, stator/magnets = 24N28P, no-load current = 0.49 A, maximum current = 54.72 A.